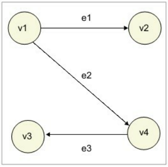

Figure 1: A simple directed graph

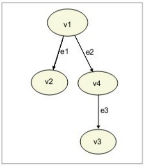

Figure 2: A DOT generated graph

```
      int num;

      printf("\nEnter an integer:");
      scanf("%d",&num);

      if( num%2==0 )
      {
        printf("%d is even number\n",num);
      }
      else
      {
        printf("%d is odd number\n",num);
      }

      return 0;
}
```

This program is so simple, it needs no explanation. We will instrument this program to automatically generate a *.dot* program file for the execution flow, based on a particular input. Shown below is the instrumented code *oddevendot.c*

```
#include<stdio.h>

int main()
{
      int num;
      FILE *fp;

      printf("\nEnter an integer:");
      scanf("%d",&num);


      if( (fp=fopen("oddeven.
dot","w"))!=NULL )
      {
        fprintf(fp,"digraph G
{\n");

        fprintf(fp,"\tstart
[label=\"%d%%2 == 0 ?\"]\n",num);
        if( num%2==0 )
        {
          fprintf(fp,"\tstart -> end [label=\"true\"]\n");
          printf("%d is even number\n",num);
          fprintf(fp,"\tend [label=\"EVEN\"]\n");
        }
        else
        {
          fprintf(fp,"\tstart -> end
[label=\"false\"]\n");
          printf("%d is odd number\n",num);
          fprintf(fp,"\tend [label=\"ODD\"]\n");
        }
        fprintf(fp,"}\n");
        fclose(fp);
      }
      else
      {
        printf("\nfopen() failed\n");
      }

      return 0;
}
```

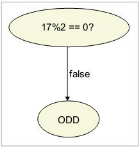

Figure 3: Execution flow graph for input 17

This program generates the dot file *oddeven.dot*. We will create the PDF graph from this file using DOT. Let's try to run, compile, execute and create the graph as shown below:

```
$ gcc oddevendot.c -o oddevendot
$ ./oddevendot
Enter an integer:17
17 is odd number
$ dot oddeven.dot -Tpdf -o oddeven.pdf
$ evince oddeven.pdf
```

Figure 3 shows the graph that will be displayed. Though this is overkill for such a simple program, it is only to demonstrate the usefulness of the programmatic way of drawing graphs.

I have only covered a very few features of the DOT language. Please explore the resources to find out how powerful and flexible the language is. You can also explore the 'neato' tool Graphviz package for drawing undirected graphs. END

References

[1] *http://www.graphviz.org/*
[2] *https://reference.wolfram.com/language/ref/format/DOT.html*

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The author works at ISRO, Sriharikota, as scientist engineer SD and has six years of industry experience. He is interested in many aspects of computer engineering and loves to explore different open source technologies.